(a) It is a platform for accelerating the transition towards food and water systems that are net-zero (carbon), nature-positive and that aims to increase farmer resilience.
(b) It is an international alliance and a network of individuals and farming organisations interested in supporting and strengthening the development of the organic animal husbandary.
(c) It is a digital platform fully integrated with service providers and built on blockchain that lets buyers, sellers and third parties trade fertilizers quickly and securely.
(d) It is a platform with the mission of encouraging the farmers to form Farmer Product Organisations or Agribusiness Consortiums, thus facilitating the access to global open markets to sell their products.

Ans.(a):
- '100 Million Farmers' is a platform that helps private and public leaders accelerate collective action to establish food and farmers as central pillars on the global climate and nature agenda and to scale up climate and nature-friendly agricultural practices.
- **Objective of 100 Million Farmers :** Build farmer resilience for food security, in support of the transition to net-Zero and nature positive food and water system by 2030.

77. Consider the following:
1. Battery storage
2. Biomass generators
3. Fuel cells
4. Rooftop solar photovoltaic units
How many of the above are considered "Distributed Energy Resources"?
(a) Only one (b) Only two
(c) Only three (d) All four

Ans.(d):
- Distributed Energy Resources (DER) refer to energy resources that are located near consumers rather than centralized power plant.
- These resources provide flexibility in electricity generation, storage, and consumption and are of various types–
  a– Rooftop solar photovoltaic units
  b–Wind power generation units
  c–Battery storage
  d–Biomass generators, which run on waste gas or industrial and agricultural by products
  e–Fuel cells.
  f–Open and closed cycle gas turbines. etc.
- Advantages of Distributed Energy Resources–
  – Improved energy efficiency
  – Flexibility and reliability
  – Incorporation of renewable energy
  – Local economic benefits.
- Distributed energy resources are playing an important role in the modern energy system and can help meet future energy demands and solve environmental problems.

78. Which one of the following shows a unique relationship with an insect that has coevolved with it and that is the only insect that can pollinate this tree?
(a) Fig (b) Mahua
(c) Sandalwood (d) Silk cotton

Ans.(a):
- Fig pollination is a unique and complex process based on the interdependence between the fig tree and the closely related fig wasps. Fig are pollinated by wasps that belong to the order Hymenoptera, family Agonidae.
- The fig fruit is actually called a syconium, an inverted flower. Its interior is suitable for pollination by the fig wasp. The female fig wasp enters the syconium of the fig and lays eggs in it. During entry, the wasp also spreads the pollen it has brought with it, which pollinates the fig flowers. The famale wasp lays her eggs in tiny flowers inside the fig. The wasp eggs hatch in to larvae, which use the flowers as food.
- Thus, this symbiotic relationship between figs and fig wasps ensures that both organisms depend on each other to survive and grow.

79. Consider the following:
1. Butterflies 2. Fish 3. Frogs
How many of the above have poisonous species among them?
(a) Only one (b) Only two
(c) All three (d) None

Ans.(c): There are some toxic species of Butterflies, fish and frogs. The main ones being the following–
Butterfly :
Monarch–Butterfly (Danaus plexippus)–This butterfly collects a toxin called cardenolide from milkweed plants in its life cycle, which makes it toxic.
Fish :
Pufferfish (Tetraodontidae family)–This fish contains a toxin called tetraodotoxin in its various organs, which is extremely lethal.
Stonefish (Synanceia)–This fish has a toxin in its spine, wich can be very painful and potentially fatal.
Frog :
Poison Dart Frogs (Dendrobatidae Family) : These frogs secrete a toxin called batrachotoxin from their skin, which is very lethal to predators.
These toxic species use the toxin to protect themselves from predators and extreme caution should be taken while handling them.

80. Consider the following:
1. Cashew 2. Papaya 3. Red sanders
How many of the above trees are actually native to India?
(a) Only one (b) Only two
(c) All three (d) None

Ans.(a):
- The native tree of India is Rakta Chandan (Pterocarpus Santalinus). Cashew (Anacardium occidentale) and papaya (Carica Papaya) are not native trees of India.
- Cashew is native to north-eastern Brazil it was brought to India by the Portuguese in the late 16th Century.
- The native place of papaya is South Mexico and Costa Rica. India is the world leader in the production of Papaya.
- The native place of red sandalwood is Andhra Pradesh.

81. Consider the following airports:
1. Donyi Polo Airport
2. Kushinagar International Airport
3. Vijayawada International Airport
In the recent past, which of the above have been constructed as Greenfield projects?
(a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

Ans.(a): Since 2019, 6 Greenfield airports namely, Kalaburagi, Orvakal (Kurnool), Shindhudurg, Itanagur, Kushinagar and Mopa have been operationalised, out of which, Kushinagar and Mopa airports are International airports.
- Donyi Polo Airport, Itanagar is the first greenfield airport has been developed in Arunachal Pradesh and was inaugurated by Prime Minister of India in 2022.

- Kushinagar International Airport, is a greenfield airport, was inaugurated by Prime Minister of India in October, 2021.
- Vijaywada International Airport is not a greenfield project.

82. With reference to "water vapour", which of the following statements is/are correct?
1. It is a gas, the amount of which decreases with altitude.
2. Its percentage is maximum at the poles.
Select the answer using the code given below:
(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

Ans.(a): Water vapour is a variable gas in the atmosphere which decreases with altitude.
- Water vapour decreases from the equator towards the poles. In the warm and wet tropics, it may account for 4% of the air by volume, while in the dry and cold areas of desert and polar regions, it may be less than 1% of the air.

83. Consider the following description:
1. Annual and daily range of temperature is low.
2. Precipitation occurs throughout the year.
3. Precipitation varies between 50cm-250 cm.
What is this type of climate?
(a) Equatorial climate
(b) China type climate
(c) Humid subtropical climate
(d) Marine West coast climate

Ans.(d): Marine west coast climate is located poleward from the Mediterranean climate on the west coast of the continents. The main areas are : North-Western Europe, west coast of North America, North of California, Southern Chile, Southern Australia and New Zealand. Due to marine influence, the temperature is moderate and in winter, it is warmer than for its latitude. The mean temperature in summer months ranges from 15° – 20° C and in winter 4°–10°C. The annual and daily ranges of temperature are small. Precipitation occurs throughout the year. Precipitation varies greatly from 50–250 cm.

84. With reference to "Coriolis force", which of the following statements is/are correct?
1. It increses with increase in wind velocity.
2. It is maximum at the poles and is absent at the equator.
Select the answer using the code given below:
(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

Ans.(c):
- The rotation of the earth about its axis affects the direction of the wind. This force is called the Coriolis Force. It deflects the wind to the right direction in the northern hemisphere and to the left in the southern hemisphere. The deflection is more when the wind velocity is high.
- The Coriolis Force is directly proportional to the angle of latitude. It is the maximum at the poles and is absent at the equator.

85. On June 21 every year, which of the following latitude(s) experience(s) a sunlight of more than 12 hours?
1. Equator 2. Tropic of Cancer
3. Tropic of Capricorn 4. Arctic Circle
Select the answer using the code given below:
(a) 1 only (b) 2 only
(c) 3 only 4 (d) 2 and 4

Ans.(d): On 21st June, the Northern Hemisphere is titled towards the sun. The rays of the sun fall directly on the Tropic of Cancer. As the result, these areas receive more neat. Sunlight of more than 12 hours is received at tropic of cancer. The North pole is inclined towards the sun and the places beyond the Arctic circle experience continuous daylight for about six months. Since a large portion of the Northern Hemisphere is getting light from the sun, it is summer in the regions north of the equator. The longest day and the shortest night at these places occur on 21st June.

86. One of the following regions has the world's largest tropical peatland, which holds about three years worth of global carbon emissions from fossil fuels; and the possible destruction of which can exert detrimental effet on the global climate. Which one of the following denotes that region?
(a) Amazon Basin (b) Congo Basin
(c) Kikori Basin (d) Rio de la Plata Basin

Ans.(b): The Congo Basin is name to the World's largest tropical peatlands. The peat swamp forest of the Congo Basin stores arround 29 billion tons of carbon-approximately equivalent to three years' Worth of global greenhouse gas emissions–while the Basin as a whole absorbs nearly 1.5 billion tons of CO_2 a year. The Basin stretches across six countries–Cameroon, Central African Republic, Democratic Republic of the Congo, Congo, Equatorial Guinea and Gabon. The Congo Basin is one of the world's last regions that absorbs more carbon than it emits. It possible destruction can have detrimental effect on the global climate.

87. With reference to perfluoroalkyl and polyfluoroalkyl substances (PFAS) that are used in making many consumer products, consider the following statements:
1. PFAS are found to be widespread in drinking water, food and food packaging materials.
2. PFAS are not easily degraded in the environment.
3. Persistent exposures to PFAS can lead to bioaccumulation in animal bodies.
Which of the statements given above are correct?
(a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

Ans.(d): Perfluroalkyl and Polyfluroalkyl substances (PEAS) are synthetic chemicals that are found in paper packaging food, drinking water, food, fabrics and non-stick cookwares.
- PFAS are not break down easily in the environment, and since they have strong bond of carbon-florine so they do not degrades easily.
- Exposure to certain PEAS may be associated with increased risk of thyroid cancer. PFAS are known to accumulate in body tissue such as in the liver.

88. Consider the following plants:
1. Carabid beetles 2. Centipedes
3. Flies 4. Termites
5. Wasps
Parasitoid species are found in how many of the above kind of organisms?
(a) Only two (b) Only three/
(c) Only four (d) All five

Ans.(b): Parasitoids are organisms that develop on or within other organism (the host), eventually killing it. Parasitoids have characteristics of both predators and parasites.
- Most parasitoids are wasps and bees (hymenoptera) or flies (Diptera) although a few species of beetles (Carabid beetles), twisted wing insects, moths have been identified as parasitoids.

89. Consider the following plants:
1. Groundnut 2. Horse-gram 3. Soybean
How many of the above belong to the pea family?
(a) Only one (b) Only two
(c) All three (d) None